

Carmen Lucca DiMario

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EDUCATION

Embry-Riddle Aeronautical University (ERAU)

Master of Science in Computer Science

Bachelor of Science in Computer Science

Graduation:

May 2027

May 2026

TECHNICAL SKILLS

Programming Languages: C, C++, Swift, GNU ASM, TypeScript, JavaScript, Java, Python, SQL, Bash

Web Frameworks: React, Next.js, Symfony, Laravel

Development Frameworks: Agile, Scrum

Tools and Technologies: UNIX, ROS (Robot Operating System), Clang, ARMcc, STM32 Microcontrollers

PROFESSIONAL EXPERIENCE

Garmin: Embedded Software Development Intern

May 2026 - Present

- Implemented Clang compiler support as a replacement for legacy ARMcc toolchains
- Extended embedded library support for 64-bit architectures, resolving data type alignment and pointer-width issues to broaden platform compatibility
- Developed a system test suite for a proprietary avionics communication interface, validating real-time engine metric transmission and ensuring protocol correctness under embedded constraints

JRC Integrated Systems: RISE Intern/CO-OP Participant

June 2025 - May 2026

- Integrated the PETSc linear algebra library into the Accuro application, reducing complex computation time
- Analyzed the advanced C/C++ codebase to identify differences in LET (Linear Energy Transfer) calculations through various substrates
- Redesigned and improved the testing framework for the entire Accuro software suite, increasing reliability

PROJECT EXPERIENCE

PFactor: iOS Application

August 2025 - Present

- Built an iOS AI oral-exam simulator for FAA checkride prep with ACS-aligned questions for 6 certificates
- Created an adaptive examiner engine scoring responses 1–5, generating targeted follow-ups on weak areas, and delivering per-category performance debriefs that mirror real DPE exam behavior
- Architected a serverless Convex backend handling auth (Sign in with Apple), StoreKit 2 subscriptions, and graceful offline degradation via automatic local-scoring fallback

F1Tenth Autonomous Racing: Software Lead

August 2025 - May 2026

- Assembled and tuned the F1Tenth hardware platform from components, including mounting compute, sensors, and motor controller into a competition-ready vehicle
- Installed and configured the full ROS2 software stack on the onboard computer
- Deployed and validated end-to-end autonomy stack from sensor input through planning and actuation, ensuring reliable performance under real competition conditions

Engineering Physics Propulsion Laboratory: AIRHOUND [Middleware Lead]

September 2024 - May 2026

- Designed a modular three-node ROS2 pipeline converting YOLOv8 detections to yaw commands via pinhole camera intrinsics, publishing to a PX4 flight controller over Micro XRCE-DDS
- Optimized YOLOv8 with NVIDIA TensorRT for embedded deployment on a Jetson Orin, enabling real-time onboard inference
- Validated full software stack via 600-second SITL test, achieving 30 Hz throughput across all pipeline stages

PUBLICATIONS

DiMario, C., Bacha, R., & Butka, B. "Combating Senior Scams Using a Large Language Model-Created Rubric." ASSE '24, ACM, 2024. <https://doi.org/10.1145/3702138.3702140>

Malarchick, R. et al. (incl. DiMario, C.) "Robust Real-Time UAV Target Tracking with Onboard Vision-Based Yaw Control." Beyond: Undergraduate Research Journal, Vol. 9, 2025. <https://commons.erau.edu/beyond/vol9/iss1/6>

EXTRACURRICULAR ACTIVITIES

- **ERAU Division II Cross-Country/Track:** 7-8 practices, approximately 30 hours, weekly
- **FAA Licenses:** Part 107 Remote Pilot Certification

<https://dimario.dev>